

1)

$$1 + \frac{8}{5} : \frac{4}{5} + \frac{5}{4} \cdot \left(\frac{7}{5} + 1\right)$$

$$1 + \frac{8}{5} \cdot \frac{5}{4} + \frac{5}{4} \left(\frac{7+5}{5}\right)$$

$$1 + 2 + \frac{5}{4} \cdot \frac{12^3}{5}$$

$$1 + 2 + 3 = 6$$

2)

$$5^{10} \cdot 5^6 : (5^8)^2$$

$$\frac{5^{(10+6)}}{5^{(8 \cdot 2)}}$$

$$\frac{5^{16}}{5^{16}} = 1$$

3)

$$\left[\left(-\frac{3}{2}\right)^4 \cdot \left(-\frac{1}{3}\right)^4 \right] \cdot \left(\frac{1}{2}\right)^{-5}$$

$$\left[\frac{(-3) \cdot (-3) \cdot (-3) \cdot (-3)}{2^4} \cdot \frac{1^4}{3^4} \right] \cdot 2^5$$

$$\left[\frac{\cancel{3^4}}{2^4} \cdot \frac{1}{\cancel{3^4}} \right] \cdot 2^5 = \frac{2^5}{2^4} = 2$$

4)

FESTEGGIATO $\frac{1}{6}$ DI TORTA

RESTANO $\frac{5}{6}$ DI TORTA PER 110 AMICI

QUINDI OGNI AMICO HA $\left(\frac{5}{6}\right) : 110 = \frac{5}{6} \cdot \frac{1}{110} = \frac{5}{660} = \frac{1}{132}$

VERIFICA 1 TORTA = $\frac{1}{6} + 110 \cdot \frac{1}{132} = \frac{1}{6} + \frac{5}{6} = 1$ CVD

5)

5	6	7	8	9
	6	7	8	
	6			
	6			

MODA = 6

$$\text{MEDIA} = \frac{5 + (6 \cdot 4) + (7 \cdot 2) + (8 \cdot 2) + 9}{10}$$

$$= \frac{68}{10} = \boxed{6,8}$$

6)

ITALIA 1,9 €/L

GERMANIA 1,4 €/L

7)

SEZIONE A $\rightarrow (165 + 89 + 146) \text{ Kg} = 400 \text{ Kg}$

SEZIONE B $\rightarrow (115 + 145 + 153) \text{ Kg} = 413 \text{ Kg}$

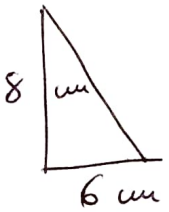
SEZIONE C $\rightarrow (158 + 137 + 92) \text{ Kg} = 387 \text{ Kg}$

A PRILE $\rightarrow (89 + 145 + 137) \text{ Kg} = 371 \text{ Kg}$

TOTALE $\rightarrow (400 + 413 + 387) \text{ Kg} = 1200 \text{ Kg}$

$$\frac{\text{SEZIONE A}}{\text{TOTALE}} = \frac{400 \text{ Kg}}{1200 \text{ Kg}} = \frac{1}{3} = 0,33... = 0,33 \approx 33\%$$

8)



Area $= \left(\frac{8 \cdot 6}{2} \right) \text{ cm}^2 = 24 \text{ cm}^2$

IPOTENUSA $= \sqrt{8^2 + 6^2} = 10 \text{ cm}$

PERIMETRO $= 6 + 8 + 10 = 24 \text{ cm}$

AREA_R $= 9 \text{ cm} \cdot 9 \text{ cm} = 36 \text{ cm}^2$

9)

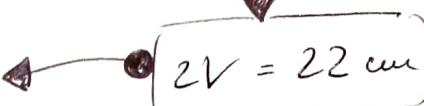
AREA_Q $= \text{AREA}_R = 36 \text{ cm}^2$

LATO_Q $= \sqrt[2]{36 \text{ cm}^2} = 6 \text{ cm}$

10)

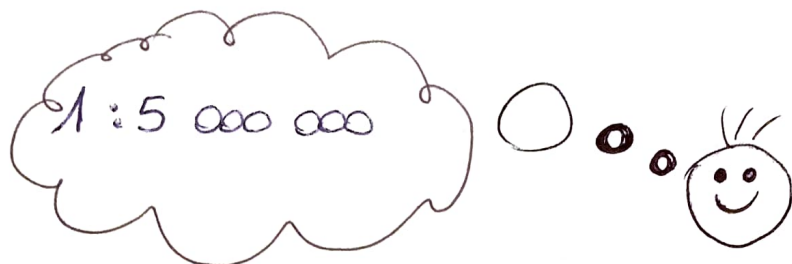
$$\text{CIRCONFERENZA} = 22\pi \text{ cm} = 2\pi r$$

a) $\pi = 3,1415\dots$

b) $r = 11 \text{ cm}$ 

c) $A = \pi r^2 = \pi \cdot (11 \text{ cm})^2 = 121\pi \text{ cm}^2 = 380,13 \text{ cm}^2$

11)



1 cm SULLA MAPPA

CORRISPONDE A ~~1000~~ 5.000.000 cm
NELLA REALTÀ

$$\begin{aligned} 22,4 \text{ cm IN MAPPA} &\Rightarrow 22,4 \cdot 5.000.000 \text{ cm} \\ &= 112.500.000 \text{ cm NELLA} \\ &\quad \text{REALTÀ} \\ &= 1.125.000 \text{ m} \\ &= 1.125 \text{ km} \end{aligned}$$

12)

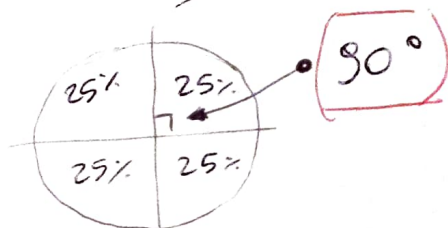
$$\underbrace{200 \text{ €} \cdot 20\%}_{\text{SCONTO}} = 200 \text{ €} \cdot \frac{20}{100} = \frac{200 \cdot 20}{100} = 40 \text{ €}$$

$$\text{PREZZO SCONTATO} = 200 \text{ €} \cdot 80\% = \frac{200 \cdot 80}{100} = 160 \text{ €}$$

13)

ALLIEVI DI VIOLINO = $200 \cdot 0,15$

ALLIEVI NO CHITARRA = $200 \cdot 0,90$



14)

$$4(2x - 3) - 5(3x - 2) = 8(1 - x)$$

$$8x - 12 - 15x + 10 = 8 - 8x$$

$$8x + 8x - 15x = 8 + 12 - 10$$

$$x = 10$$